

SATADEEP DASGUPTA

Solution Architect & Systems Engineer

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> Professional Summary

Software engineer specializing in execution runtimes, workflow orchestration systems, and developer infrastructure with 5+ years of experience building production-grade distributed systems. Currently working as a Solution Architect at Brainium Information Technologies, where I design scalable application architectures and distributed systems. Built Loom, a durable workflow execution engine that enables deterministic execution and event-sourced replay for long-running processes, and AIME, an AI-powered adaptive learning platform presented to national leadership. Published researcher with expertise in hybrid database architectures for RAG systems. Proven track record in systems programming, infrastructure tooling, and backend platform development using Rust, C++, Python, and TypeScript.

> Experience

Solution Architect

[Brainium Information Technologies Pvt Ltd](#)

June 2024 – Present

Solution Architect (January 2025 – Present)

- ▶ Lead architecture design for enterprise software systems and cloud-native applications serving 50,000+ users
- ▶ Design scalable backend architectures and distributed workflows with 99.9% uptime requirements
- ▶ Develop AI-assisted systems including AIME (Adaptive Instruction & Misconception Engine), presented to national leadership and 14 state education inspectors
- ▶ Built Loom workflow orchestration engine with deterministic execution and event-sourced replay capabilities
- ▶ Guide system design decisions including service decomposition, data architecture, and observability strategies for mission-critical applications

Associate Software Engineer (June 2024 – January 2025)

- ▶ Implemented core system components for enterprise applications handling 100K+ daily transactions
- ▶ Solved complex backend engineering problems involving integrations and data pipelines, reducing processing time by 40%
- ▶ Worked on distributed application components and performance-sensitive backend services with sub-100ms response times
- ▶ Contributed to codebase modernization efforts improving maintainability and reducing technical debt

Full Stack Developer

[IGLOBAL IMPACTS ITES Pvt Ltd](#)

June 2022 – June 2024

- ▶ Designed and maintained scalable web applications and backend services supporting 10,000+ concurrent users

- Built and maintained CI/CD pipelines reducing deployment time by 50% and deployment errors by 90%
- Led software releases improving application performance by 40% and reducing page load times by 30%
- Participated in technical hiring and engineering evaluation processes, conducting 30+ technical interviews
- Implemented production-grade RESTful APIs and GraphQL services with 99.5% uptime and sub-200ms response times
- Mentored junior developers on best practices in code versioning, testing, and system design

Freelance Software Developer

August 2019 – June 2022

- Delivered 25+ end-to-end web application projects for international clients across 8 countries
- Designed full-stack systems from requirement gathering to deployment with 100% on-time delivery rate
- Built scalable backend APIs serving 50K+ requests per day with optimized frontend achieving 95+ Lighthouse scores
- Maintained production deployments and resolved complex technical issues with average resolution time under 4 hours
- Achieved 5.0/5.0 client satisfaction rating with multiple recurring clients

> Technical Skills

Languages

Rust C++ Python TypeScript JavaScript Java

Backend

Node.js Django Express.js REST APIs GraphQL

Frontend

React Next.js Vue.js Tailwind

Databases

PostgreSQL MongoDB MySQL Redis SurrealDB

DevOps & Tools

Docker AWS CI/CD Git GitHub Actions Jenkins

Specialized

Event Sourcing Workflow Orchestration Distributed Systems Systems Programming

> Key Projects

AIME — Adaptive Instruction & Misconception Engine

- Generates structured lessons dynamically from curriculum inputs
- Captures learner interaction evidence to detect misconceptions and adapt lesson flow
- Introduced Lesson Runtime Format (LRF) for replayable lesson execution
- Presented to national leadership and 14 state education inspectors with positive reception

AI Systems, Workflow Orchestration, Python

Loom — Durable Workflow Execution Engine

- Implements event-sourced workflow execution and deterministic replay

- Supports fault-tolerant workflow execution with state reconstruction from event logs
- Designed for distributed systems requiring reliable orchestration of asynchronous tasks

[Python](#), [Event Sourcing](#), [Distributed Systems](#)

Ironwatch — DDoS Mitigation System

- Detects abnormal traffic spikes using threshold-based monitoring
- Uses AbuseIPDB reputation data to evaluate malicious IPs
- Reduced attack traffic by approximately 80% in production deployment

[Rust](#), [Network Monitoring](#), [Security](#)

OxideDB — LLM-Powered Database System

- Dual query modes supporting SQL and natural language queries via LLM integration
- High-performance gRPC protocol with Protocol Buffers for binary communication
- Knex.js-inspired SDKs for Rust, Python, and JavaScript/TypeScript
- Multi-tenant architecture with namespace isolation and JWT authentication
- Full SQLite support including CTEs, window functions, and triggers

[Rust](#), [SQLite](#), [LLM](#), [gRPC](#), [Protocol Buffers](#)

cvxtract — Schema-Driven Document Extraction

- Document processing pipeline supporting PDF, DOCX, HTML, and TXT inputs
- Typed structured extraction using Serde and JSON Schema for compile-time validation
- Provider-agnostic LLM integration with local inference (llama.cpp) and cloud API support
- Custom schema extraction enabling direct deserialization into Rust structs
- Modular architecture with document loaders, extraction pipeline, and model abstraction

[Rust](#), [LLM](#), [Document Processing](#), [Serde](#), [JSON Schema](#)

> Key Achievements

- ★ **Published Research:** Co-authored peer-reviewed research on hybrid database architectures for RAG systems
- ★ **Production Impact:** Deployed DDoS mitigation system reducing attack traffic by 80%
- ★ **Leadership Recognition:** Presented AIME platform to national leadership and 14 state education inspectors
- ★ **Career Progression:** Advanced from Freelancer → Full Stack → Solution Architect in 5 years
- ★ **Performance Excellence:** Delivered systems with 99.9% uptime and sub-100ms latency
- ★ **Open Source:** Maintain 15+ active repositories with global contributions

> Research & Publications

Bridging Analytics and Semantics: A Hybrid Database Approach to Retrieval-Augmented Generation

Co-author | Zenodo Repository, September 2025

DOI: [10.5281/zenodo.17018699](https://doi.org/10.5281/zenodo.17018699) | Source: github.com/satadeep3927/hybridrag

- Combines vector search with semantic reasoning for improved retrieval accuracy
- Demonstrated 18% improvement in relevance accuracy over vector-only baselines

> Education

Bachelor of Science (Mathematics)

Brahmananda Keshab Chandra College

2019 – 2022

> Languages

- English (Fluent)
- Bengali (Native)
- Hindi (Fluent)